

Arithmophysics: Decay Inside the Growth

Complete continuation prompt: parity alignment, prime packets, and finite descent

Christophe Michaels research programme

October 6, 2026

Directive

Continue this research from the checkpoint below. Investigate Christophe Michaels's proposal that the remaining subpower growth contains a measurable decay: the increasingly precise cancellation between even and odd prime-factor colors. Determine whether its behavior across scales supplies a contracting recurrence. Treat apparent fractal or self-similar structure as a mathematical hypothesis to formulate and test.

The immediate task is to **prove an independent upper bound for one specified signed arithmetic remainder**, then carry it back to the original energy. Prioritize the completed-prime-packet remainder in Section 4, using parity alignment to identify the required size. A bound restricted to the retained records in Section 5 is sufficient. The alternative investigations are tools for this task, not additional prerequisites.

Execute the proof attempts and the calculations. Preserve useful partial inequalities. Distinguish exact identities, uniform estimates, and finite diagnostics. An equivalent sufficient condition is a valid research target; the proof must obtain its control from stated arithmetic inputs rather than assume that target under another name. State this distinction once and proceed with the mathematics.

Each new representation must identify its forward map, its return map or identity, its costs, and the specific estimate it makes more accessible. Complete a full return within this investigation. If an attempt fails, identify the exact surviving term and the strongest inequality obtained. A representation by zeros, or an algebraic collapse, does not by itself exhaust other possible estimates.

1. Original energy and the fixed compression

For integers $N \geq 1$, set

$$M(x) = \sum_{n \leq x} \mu(n), \quad c_N(n) = \frac{\mu(n)}{n}, \quad B_N(a, b) = \min(a, b).$$

All logarithms are natural. Nonintegral summation limits mean their floors. Write $\|v\|_B^2 = v^* B_N v$. The complete energy is

$$R(N) = \|c_N\|_B^2 = \sum_{a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)} = \sum_{k < N} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}. \quad (1)$$

The terminal term remains part of every return.

Use this exact mesh, independently of the source:

$$x_0 = 0, \quad x_{r+1} = \min\{N, x_r + \max(1, \lfloor N^{-1/6} x_r^{2/3} \rfloor)\}, \quad x_m = N. \quad (2)$$

Compute the step with the integer sixth root of $\lfloor x_r^4/N \rfloor$ to avoid floating-point boundary changes. There are at most $13\sqrt[6]{N} + 2$ intervals.

For an interval $(a, b]$, the projection $\mathcal{T}_N$ replaces a coefficient basis vector by

$$\mathcal{T}_N e_n = \frac{b-n}{b-a} e_a + \frac{n-a}{b-a} e_b, \quad a < n \leq b, \quad (3)$$

where $e_0 = 0$. The first interval is the unchanged singleton $(0, 1]$. Add contributions at shared boundaries. Equivalently, for physical charges a_n , define

$$Q = \sum_{a < n \leq b} a_n, \quad H = \sum_{a < n \leq b} \frac{a_n}{n}.$$

For $a > 0$, the two boundary coefficients are $(bH - Q)/(b - a)$ and $(Q - aH)/(b - a)$. They preserve both coefficient mass and physical charge.

This is the B_N -orthogonal projection onto the boundary span. For the Möbius source, put $p_N = \mathcal{T}_N c_N$. The proved return is

$$R(N) = E_{\mathcal{P}}(N) + E_{\text{res}}(N), \quad E_{\mathcal{P}} = \|p_N\|_B^2, \quad 0 \leq E_{\text{res}} \leq C_0 := 2^{-4/3}. \quad (4)$$

The error bound holds for every real physical source $|a_n| \leq 1$. For

$$u_N(x) = \sum_{n \leq N} \frac{\mu(n)}{n} \min(x, n),$$

the retained energy and its terminal lower bound are

$$E_{\mathcal{P}}(N) = \sum_{r=1}^m \frac{[u_N(x_r) - u_N(x_{r-1})]^2}{x_r - x_{r-1}} \geq \frac{M(N)^2}{N}. \quad (5)$$

On $(a, b]$, the mean slope is

$$\sum_{n > b} \frac{\mu(n)}{n} + \frac{1}{b-a} \sum_{a < n \leq b} \mu(n) \frac{n-a}{n}. \quad (6)$$

Keep these inherited and local contributions together inside the square.

The direct target is $E_{\mathcal{P}}(N) \ll_{\varepsilon} N^{\varepsilon}$ for every $\varepsilon > 0$. Section 5 gives a weaker sufficient target at selected records.

Reconcile the projection before new comparisons. For this mesh and projection, at $N = 16384$,

$$m = 382, \quad R = 1.6034402209313143, \quad E_{\mathcal{P}} = 1.5883431079039165, \quad E_{\text{res}} = 0.015097113027358108.$$

A report using a different split, differing by about 1.5×10^{-3} , must specify its different map or indexing. A fixed orthogonal projection is unique; agreement of the total alone does not reconcile the components. The bound 3.3 reported elsewhere is not the sharp proved cap in (4).

2. The decay variable: relative parity defect

Let $\omega(n)$ count distinct prime factors, with $\omega(1) = 0$. Define

$$c_N^{[k]}(n) = \frac{\mu(n)^2}{n} \mathbf{1}_{\{\omega(n)=k\}}, \quad p_k = \mathcal{T}_N c_N^{[k]}.$$

All colors use the same projection. Set

$$e_N = \sum_{k \text{ even}} p_k, \quad o_N = \sum_{k \text{ odd}} p_k, \quad p_N = e_N - o_N.$$

Define the parity background and its relative defect by

$$\mathcal{A}_N = \|e_N\|_B^2 + \|o_N\|_B^2, \quad \delta_N = \frac{E_{\mathcal{P}}(N)}{\mathcal{A}_N}. \quad (7)$$

This $\mathcal{A}_N$ is not the Ramanujan-carrier input energy from the earlier paper.

The normalization has an unconditional linear scale. Put $c_{\text{sf}} = 2 - \pi^2/6 > 0$. Then

$$\frac{c_{\text{sf}}^2}{2} N \leq \mathcal{A}_N \leq 2N, \quad 0 \leq \delta_N \leq 1. \quad (8)$$

Here is a short proof to verify and reuse. The number Q_N of squarefree integers through N is at least $c_{\text{sf}}N$, by the union bound over square divisors d^2 , $d \geq 2$. The positive squarefree source has retained energy $U_N = \|e_N + o_N\|_B^2 \geq Q_N^2/N$. Its full energy is at most the all-positive energy $2N - H_N \leq 2N$, where $H_N = \sum_{n \leq N} 1/n$. Projection decreases energy. Since the boundary coefficients and Green entries are nonnegative, $\langle e_N, o_N \rangle_B \geq 0$, giving $U_N/2 \leq \mathcal{A}_N \leq U_N$ and (8).

Consequently the decay target is precisely

$$\boxed{\delta_N \ll_{\varepsilon} N^{-1+\varepsilon}} \iff E_{\mathcal{P}}(N) \ll_{\varepsilon} N^{\varepsilon} \quad \text{for every } \varepsilon > 0. \quad (9)$$

This makes “decay inside the growth” quantitative: a relative mismatch decays against a background of order N , while its absolute energy may still grow slowly or fluctuate. Monotonic decay, a limiting constant, and a uniform C/N bound are not required.

For $N \geq 2$, let $a_N = \|e_N\|_B$, $b_N = \|o_N\|_B$, and $\cos \theta_N = \langle e_N, o_N \rangle_B / (a_N b_N)$. Then

$$E_{\mathcal{P}} = (a_N - b_N)^2 + 2a_N b_N (1 - \cos \theta_N). \quad (10)$$

These are the amplitude mismatch and directional mismatch. Control their sum; numerical dominance of the second does not prove the first harmless.

The following finite diagnostics were computed with (2)–(5):

N	$E_{\mathcal{P}}$	$\mathcal{A}_N$	δ_N	$N\delta_N$
1024	1.444694559	383.495976	0.00376717006	3.857582
16384	1.588343108	6061.860996	0.000262022357	4.292974
65536	1.579293249	24229.290027	0.0000651811608	4.271713
262144	1.613459198	96886.510648	0.0000166530840	4.365506
1000000	1.702649419	369583.407438	0.00000460694226	4.606942

The directional term contributes about 95.5 percent of (10) at 16384 and 99.1 percent at 10^6 . These figures motivate the investigation; they establish no asymptotic exponent.

Compute small differences from the signed source directly. In the Euclidean slope representation $v_r = \Delta u / \sqrt{\Delta x}$, use $v_{\mu} = v_e - v_o$ and

$$2a_N b_N (1 - \cos \theta_N) = a_N b_N \left\| \frac{v_e}{a_N} - \frac{v_o}{b_N} \right\|_2^2.$$

Avoid obtaining the entire residual by subtracting two large rounded energies.

3. What the color calculation has already located

The exact Gram matrix is $G_{k\ell} = \langle p_k, p_\ell \rangle_B$. If q_k counts squarefree integers with $\omega(n) = k$, and V_{rk} are their projected Euclidean slopes, then

$$G = \frac{qq^*}{N} + C, \quad C = \left(V - \frac{wq^*}{N}\right)^* \left(V - \frac{wq^*}{N}\right) \succeq 0, \quad w_r = \sqrt{x_r - x_{r-1}}. \quad (11)$$

For the parity vector $s_k = (-1)^k$,

$$E_{\mathcal{P}} = \frac{M(N)^2}{N} + s^* C s. \quad (12)$$

Near rank one is therefore a description of the large coherent component. The centered parity energy and the terminal still need arithmetic bounds.

At $N = 16384$, the same-color sum is 3845.369090532, the signed mixed sum is -3843.780747424 , and their total is 1.588343108. Refinement by small-prime incidence changes the diagonal decomposition while preserving the total. A smaller diagonal after regrouping is not itself a gain.

Prime insertion supplies an exact relation, with every truncation retained:

$$k p_k = \sum_{p \leq N} \mathcal{T}_N \mathcal{D}_{p,N} c_{p \cdot}^{[k-1]}, \quad \mathcal{D}_{d,N} e_m = d^{-1} e_{dm} \mathbf{1}_{\{dm \leq N\}}. \quad (13)$$

The child source in (13) is the factor-count source restricted to $p \nmid m$. Projection and dilation are not silently commuted. Use this relation if it gives a compulsory estimate for (10) or (12), with its full sum of costs.

4. Priority attempt: the completed-packet remainder

Let $\mathcal{C}$ be a nonempty finite prime set and $q = \prod_{p \in \mathcal{C}} p$. The exact finite identity is

$$c_N = \prod_{p \in \mathcal{C}} (I - \mathcal{D}_{p,N}) f_{\mathcal{C},N}, \quad f_{\mathcal{C},N}(m) = \frac{\mu(m)}{m} \mathbf{1}_{\{(m,q)=1\}}. \quad (14)$$

Define g_N using only parents $m \leq N/q$, with all their descendants:

$$g_N = \sum_{\substack{m \leq N/q \\ (m,q)=1}} \sum_{d|q} \frac{\mu(m)\mu(d)}{dm} e_{dm}, \quad t_N = c_N - g_N. \quad (15)$$

Thus t_N includes every unfinished parent with its inherited cutoff. Complete packets have zero total physical charge; the terminal lies in t_N . Their cross term remains.

Put

$$P_{\mathcal{C}}(y) = \sum_{d|q} \mu(d) \mathbf{1}_{\{d \leq y\}}, \quad K_{\mathcal{C}}(m, n) = \sum_{d, e|q} \frac{\mu(d)\mu(e)}{\max(dm, en)}.$$

The already established kernel formula is

$$K_{\mathcal{C}}(m, n) = \int_0^\infty P_{\mathcal{C}}(x/m) P_{\mathcal{C}}(x/n) \frac{dx}{x^2}. \quad (16)$$

It vanishes for $\max(m, n) \geq q \min(m, n)$, and

$$K_{\mathcal{C}}(m, m) = \frac{\kappa_{\mathcal{C}}}{m}, \quad \kappa_{\{2,3\}} = \frac{2}{3}, \quad \kappa_{\{2,3,5\}} = \frac{13}{15}.$$

Start with $\mathcal{C} = \{2, 3\}$. Define the **whole remaining signed term**

$$\begin{aligned} \mathcal{I}_{\mathcal{C}}(N) = & 2 \sum_{\substack{m < n \leq N/q \\ (mn, q) = 1, \, n < qm}} \mu(m)\mu(n)K_{\mathcal{C}}(m, n) \\ & + 2\langle g_N, t_N \rangle_B + \|t_N\|_B^2. \end{aligned} \quad (17)$$

The exact return is

$$R(N) = \kappa_{\mathcal{C}} \sum_{\substack{m \leq N/q \\ (m, q) = 1}} \frac{\mu(m)^2}{m} + \mathcal{I}_{\mathcal{C}}(N). \quad (18)$$

The diagonal costs at most $\kappa_{\mathcal{C}}(1 + \log N)$. It is already affordable. Estimate the joint remainder without spending its cross-term cancellation.

At $N = 16384$, for $\{2, 3\}$, the full-metric components are: diagonal 2.053503, local overlap -0.537728 , completed–unfinished cross -0.024996 , and unfinished square 0.112661. Their sum is $R(N)$, not $E_{\mathcal{P}}(N)$.

A concrete passage to logarithmic coordinates

Derive and check the normalization of this candidate representation before using it:

$$\begin{aligned} K_{\mathcal{C}}(m, n) &= \frac{1}{\sqrt{mn}} k_{\mathcal{C}}(\log(n/m)), \\ \widehat{k}_{\mathcal{C}}(\xi) &= \frac{\prod_{p \in \mathcal{C}} |1 - p^{-1/2 + i\xi}|^2}{\xi^2 + 1/4}, \quad \widehat{k}(\xi) = \int_{\mathbb{R}} k(u) e^{-i\xi u} du. \end{aligned} \quad (19)$$

This follows by finite differences of $e^{-|u|/2}$; $k_{\mathcal{C}}$ has support in $[-\log q, \log q]$. It exposes a finite-range overlap in log scale and a prime filter in frequency. Equivalently, the original field map is

$$\Phi_c(v) = \sum_{n \leq N} \sqrt{n} c_n e^{-|v - \log n|/2}, \quad \int_{\mathbb{R}} \left(|\Phi'_c|^2 + \frac{1}{4} |\Phi_c|^2 \right) dv = \|c\|_B^2. \quad (20)$$

For a fixed prime set, the filter in (19) lies between $\prod_p (1 - p^{-1/2})^2$ and $\prod_p (1 + p^{-1/2})^2$. Thus a generic operator norm gives fixed comparability, not an arithmetic power saving. The opportunity is the actual coprime Möbius source and its joint interaction with t_N . Attempt an estimate exploiting that structure. If $\mathcal{C}$ grows with N , account explicitly for q , support, unfinished parents, and every prime-dependent constant.

Universal contraction for an individual packet is already disproved. For $\mathcal{C} = \{2, 3\}$, the coprime parent with charges $a_1 = 1, a_5 = -1$, completed at $N = 30$, has parent energy $4/5$, completed energy $13/15$, and compressed completed energy $4637/5400 > 4/5$. Use a collective or arithmetic-family estimate instead.

5. A finite return that would settle every further descent

Let

$$\mathcal{S}(X) = \max_{0 \leq n \leq X} R(n), \quad R(0) = 0, \quad \tau_h = \min\{n : |M(n)| = h\}.$$

Every strict energy record is an absolute-Mertens first passage. At a noninitial record N , write $h = |M(N)|$ and $a = \tau_{h-1}$. This a is the preceding adjacent-level first passage; it need not be an energy record.

Retain the initial record and those satisfying

$$N - a < \frac{\sqrt{N}}{[1 + \mathcal{S}(N^{1/6})]^{5/2}}. \quad (21)$$

Let $T_{\text{ret}}(X)$ be the maximum of 1 and $M(N)^2/N$ on these retained records through X . The supplied record-accounting theorem is

$$\mathcal{S}(X) \leq (4 \log X + 4)T_{\text{ret}}(X) + 2(\log X)^2[1 + \mathcal{S}(X^{1/6})]^5. \quad (22)$$

The underlying exact record identity is

$$R(N) - R(a) = \frac{2h - 1}{N} - \sum_{k=a}^{N-1} \frac{(h - 1)^2 - M(k)^2}{k(k + 1)}. \quad (23)$$

For completeness, the accounting proof is included here. If a previous index $v < N$ attained the maximum absolute height H through N , then (1) gives

$$R(N) - R(v) \leq \frac{M(N)^2 - H^2}{N} \leq 0.$$

Thus a strict record is a new first passage, and different records have distinct heights. At a record of height h , the energy-envelope jump is at most $(2h - 1)/N$, and its square-root jump is at most $N^{-1/2}$. For the latter, write the two consecutive energies as a common nonnegative historical term plus h^2/N and $(h - 1)^2/N$.

The sum $A_{\text{ret}}(X)$ of ordinary retained jumps therefore satisfies

$$A_{\text{ret}}(X) \leq T_{\text{ret}}(X) \sum_{h \leq X} \frac{2h - 1}{h^2} \leq (2 \log X + 2)T_{\text{ret}}(X).$$

Write $\mathcal{H}(X) = [1 + \mathcal{S}(X^{1/6})]^{5/2}$. For every discarded record, $\Delta\sqrt{\mathcal{S}}(N) \leq N^{-1/2} \leq \mathcal{H}(N)(N - a)/N$. The adjacent-height first-passage intervals $(a, N]$ are disjoint. Consequently

$$\sum_{\text{discarded } N \leq X} \Delta\sqrt{\mathcal{S}}(N) \leq \mathcal{H}(X) \int_1^X \frac{dt}{t} = \mathcal{H}(X) \log X.$$

At each retained jump, the square-root increment of $\mathcal{S}$ is at most the square-root increment of A_{ret} , because $\mathcal{S} \geq A_{\text{ret}}$. Summing both classes yields $\sqrt{\mathcal{S}(X)} \leq \sqrt{A_{\text{ret}}(X)} + \mathcal{H}(X) \log X$. Squaring with $(u + v)^2 \leq 2u^2 + 2v^2$ proves (22). No earlier manuscript is needed to use this return.

Any one of the following estimates, for every $\varepsilon > 0$ and only on retained records, supplies a sufficient arithmetic input:

$$\begin{aligned} \delta_N &\leq C_\varepsilon N^{-1+\varepsilon} [1 + \mathcal{S}(N^{1/6})]^5, \quad \text{or} \\ [\mathcal{I}_{\{2,3\}}(N)]_+ &\leq C_\varepsilon N^\varepsilon [1 + \mathcal{S}(N^{1/6})]^5, \quad \text{or} \\ \frac{M(N)^2}{N} &\leq C_\varepsilon N^\varepsilon [1 + \mathcal{S}(N^{1/6})]^5. \end{aligned} \quad (24)$$

Here $[x]_+ = \max(x, 0)$. These are alternative sufficient targets, not three obligations and not assertions that the energy targets follow from the terminal target. The first uses (8) and (5); the second uses (18) and (1). Absorb the logarithmic diagonal and the logarithms in (22) by applying (24) with a smaller exponent.

The resulting recurrence is

$$1 + \mathcal{S}(X) \leq C_\eta X^\eta [1 + \mathcal{S}(X^{1/6})]^\eta \quad \text{for every } \eta > 0. \quad (25)$$

This closes because $5/6 < 1$. More generally the criterion is

$$1 + \mathcal{S}(X) \leq C_\eta X^\eta [1 + \mathcal{S}(X^\beta)]^\gamma, \quad 0 < \beta < 1, \quad \gamma > 0, \quad \gamma\beta < 1. \quad (26)$$

The elementary bound $\mathcal{S}(X) \leq 2X$ makes $\rho = \limsup \log(1 + \mathcal{S}(X))/\log X$ finite. Equation (26) gives $\rho \leq \eta + \gamma\beta\rho$; letting $\eta \downarrow 0$ gives $\rho = 0$.

Also show the finite iteration explicitly. Put $F(t) = \log(1 + \mathcal{S}(e^t))$. For (25),

$$F(t) \leq 5F(t/6) + \eta t + \log C_\eta.$$

After descending to a fixed base scale, there are $O(\log \log X)$ steps, and

$$F(\log X) \leq 6\eta \log X + O_\eta((\log X)^{\log 5/\log 6}). \quad (27)$$

The second exponent is less than one. This is how a valid scale contraction controls the entire repeating structure in one argument. A fixed power loss X^κ , $\kappa > 0$, would instead leave a positive exponent; identify such a loss if it appears.

For a reconstruction at the sixth-root scale, let $K = \lfloor N^{1/6} \rfloor$, $A = \mu \mathbf{1}_{[1,K]}$, and let $*$ denote Dirichlet convolution. Then

$$\mu(n) = \sum_{j=1}^6 (-1)^{j-1} \binom{6}{j} (A^{*j} * \mathbf{1}^{*(j-1)})(n), \quad n \leq N, \quad (28)$$

with $\mathbf{1}(n) = 1$ and zeroth convolution power equal to the convolution identity. This follows because $(\delta_1 - A * \mathbf{1})^{*6}$ is supported above N . Keep all six signed terms together. Seek a proved exponent $\gamma < 6$, rather than inferring it from the number of factors.

At $N = 16384$, their separate energy sum is 5564.983833, while their complete signed energy is 1.588343108. The previous record scan through 16384 found records at 1, 5, 13, 31, with only the initial one retained. Such data do not test (24) at a noninitial retained record.

6. Two supporting investigations, selected by what the main attempt needs

A. Resolve the parity-phase profile with an absolute remainder

Use the finite vector polynomial

$$\mathcal{F}_N(z) = \sum_k z^k p_k, \quad \mathcal{E}_N(t) = \|\mathcal{F}_N(e^{it})\|_B^2.$$

Its parity value is $\mathcal{E}_N(\pi) = E_{\mathcal{P}}(N)$. Its angular average is $\sum_k G_{kk}$. The corresponding full-energy profile has value $R(N)$ at parity; label the two separately.

The Dirichlet generating function is, initially for $\Re s > 1$,

$$\sum_{n \geq 1} \frac{\mu(n)^2 z^{\omega(n)}}{n^s} = \prod_p (1 + zp^{-s}) = \zeta(s)^z H(s, z), \quad H(s, z) = \prod_p (1 + zp^{-s})(1 - p^{-s})^z. \quad (29)$$

At $z = -1$, $H(s, -1) = 1$. Under the relevant Selberg–Delange hypotheses, expansion coefficients contain $1/\Gamma(z - j)$; at $z = -1$ all those algebraic coefficients vanish. This locates an endpoint remainder. It does not estimate that remainder at the required scale. A source for the coefficient formula is de

la Bretèche and Tenenbaum, *Remarks on the Selberg–Delange method*, equations (1.9)–(1.10): <https://arxiv.org/pdf/2010.12929>.

Derive an **absolute** estimate uniform in a stated shrinking interval $|t - \pi| \leq w_N$, and propagate it through the finite projection with its cost. State the hypotheses of the theorem used. Relative agreement with a main term that vanishes at parity cannot control the endpoint error.

An exact finite diagnostic is available without an asymptotic theorem. Define $H_1 = \sum_k k(-1)^k p_k$ and $H_2 = \sum_k k^2(-1)^k p_k$. Then

$$\mathcal{E}'_N(\pi) = 0, \quad \mathcal{E}''_N(\pi) = 2\|H_1\|_B^2 - 2\langle p_N, H_2 \rangle_B. \quad (30)$$

Measure the curvature, amplitude defect, directional defect, and profile width on the same mesh. Do not assume the curvature is positive or that parity is a global minimum. If a repeated scaled profile emerges, formulate its candidate scaling law with an error term and test fresh horizons. A profile fit supplies a conjectured inequality, not its uniform proof.

B. Separate the shuffle variance from the arithmetic ordering

For squarefree positions I_j in $[2^j, 2^{j+1}) \cap [1, N]$, let $k_j = |I_j|$ and $s_j = \sum_{n \in I_j} \mu(n)$. Shuffle the existing signs uniformly within each block, independently across blocks, leaving zeros fixed. The mean source is $\bar{a}_n = s_j/k_j$ on I_j . Empty blocks are omitted.

For the physical-charge kernel

$$\widehat{K}_{nm} = \frac{\langle \mathcal{T}_N e_n, \mathcal{T}_N e_m \rangle_B}{nm},$$

the exact expectation is

$$\begin{aligned} \mathbb{E}E_{\mathcal{P}}(\mu_{\text{shuffle}}) &= E_{\mathcal{P}}(\bar{a}) + V_N, \\ V_N &= \sum_{j:k_j \geq 2} \frac{1 - (s_j/k_j)^2}{k_j - 1} \left[k_j \operatorname{tr} \widehat{K}_{I_j} - \mathbf{1}^* \widehat{K}_{I_j} \mathbf{1} \right], \quad 0 \leq V_N \leq 2D_N, \end{aligned} \quad (31)$$

where $D_N = \sum_{n \leq N} \mu(n)^2/n \leq 1 + \log N$. Define

$$\Delta_N = E_{\mathcal{P}}(\mu) - \mathbb{E}E_{\mathcal{P}}(\mu_{\text{shuffle}}).$$

Thus the remaining signed combination is $E_{\mathcal{P}}(\bar{a}) + \Delta_N$, with a paid logarithmic variance. If this route is useful, bound that combination directly. An assumed sign for Δ_N is insufficient unless the mean-source energy is also controlled.

At $N = 16384$, the exact expectation is 2.035225552, comprising mean energy 1.473291335 and variance 0.561934217. The actual energy is 1.588343108. A block-conditioned shuffle and independent random signs are distinct controls.

7. Calculations and controls that answer the question

Use the existing scripts and data when available. Independently reproduce only the checkpoint needed to establish common definitions, then spend the computation on the proposed inequality.

1. Report $E_{\mathcal{P}}, E_{\text{res}}, \mathcal{A}_N, \delta_N, N\delta_N$, both terms of (10), and the terminal. Compute envelope and local scale exponents without assuming monotonicity. Extend beyond the existing table if resources permit; include intermediate horizons and local peaks, not only powers of ten.
2. Compute every term of (17), using the full metric, and also its compressed version if that is used. Identify which joint cancellation survives passage between scales. Test the exact candidate bound suggested by the proof attempt, rather than only plotting its components.

3. If pursuing (24), scan actual energy records and retained records, with numerical uncertainty controlled near ties. Report an empty noninitial retained set explicitly. Ordinary horizons are not substitutes for that test.
4. Use all-positive charges, squarefree-positive charges, independent random signs on square-free support, and the exact conditional shuffle. For finite prime colors also use $a_{\mathcal{C}}(n) = \mu(n)^2(-1)^{\#\{p \in \mathcal{C}: p|n\}}$. For fixed $\mathcal{C}$, this last source has linear energy with leading coefficient $2\alpha_{\mathcal{C}}^2$, where $\alpha_{\mathcal{C}} = (6/\pi^2) \prod_{p \in \mathcal{C}} (p-1)/(p+1) > 0$. Its fixed-set error constants are not uniform for a growing prime set.
5. State precisely what a control tests. Independent random signs can already have logarithmic energy, so separating Möbius from that control is not necessary for a subpower bound. The coherent positive controls test whether an alleged estimate improperly ignores arithmetic signs.

Use prefix sums and the slope formula for linear-time energy calculations. Avoid constructing an $N \times N$ matrix at large N . Use exact rational checks at small sizes and compensated or higher-precision checks when large signed components nearly cancel. Keep errors in energies distinct from errors in their square roots.

8. Cost accounting and the required outcome

For the principal attempt, supply a short table listing each transformation, the exact preserved object, the estimate used, and the full dependence of its cost on N , prime cutoff, phase width, or child scale. The known compression cost is additive C_0 . The known packet diagonal and shuffle variance cost only logarithms. Identify any step that instead loses a fixed power.

The earlier bounded inverse for the log-weighted source is a transfer estimate. Its observed full norm near 25.45 and the tail norm near 0.532 at cutoff 21 are finite numerical values, not new uniform arithmetic estimates. Re-evaluating that identity is unnecessary unless the current proof uses a genuinely new inequality for its input.

Likewise, enlarging a positive carrier space requires controlling both the observation norm and the size of the input placed there. The earlier Ramanujan observer had logarithmic observation cost but input energy of order N . A new lift should show exactly where its quantitative gain enters.

Return the following, in this order:

1. **Strongest new proved inequality.** State its range, constants, quantifiers, and complete proof. List the independent arithmetic inputs actually used. If no new arithmetic inequality is obtained, give the strongest rigorous partial result first.
2. **Complete return.** Carry that inequality through (4), (8), (18), or (22), as appropriate. Display the resulting growth exponent or the contraction parameter γ/β , including all costs and endpoints. If it closes, finish the proof in the original energy.
3. **Growth-and-decay finding.** State whether the evidence supports a useful scaling law, with what error, and whether it yields a recurrence. Separate relative parity decay, absolute energy growth, and phase width.
4. **One reproducible computation package.** Supply the script, data, figures, exact checkpoints, control definitions, and a note distinguishing measured numbers from uniform bounds.
5. **If the target remains:** display one fully specified surviving expression and the exact inequality still needed. Explain which tested estimate failed and at which cost. Formulate the next attempt from that calculation, without treating an untested representation as exhausted.

The aim of this continuation is a finite, reviewable advance: an arithmetic estimate and its return, a weaker contraction that still closes, or a sharply quantified partial inequality that materially narrows the remaining task. Repeated structure is useful only insofar as its total cost can be controlled.